

Bandon Public Transport Accessibility Analysis

Bandon, County Cork, Republic of Ireland

SEL-002 | Public Transport Accessibility & Vulnerability

Prepared as a reproducible GIS-based spatial evidence study

September 2026



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Abstract

Bandon is one of the fastest-growing towns in West Cork and occupies a distinctive position within the regional settlement hierarchy. According to Census 2022, the town has a resident population of 8,196 and functions simultaneously as a commuter town within the Greater Cork metropolitan area and as a gateway to the wider West Cork region. Its strategic location approximately 30 km southwest of Cork City makes it an important origin and destination for daily commuting, education, healthcare, retail, and regional services.

This dual role raises critical questions about the performance of the public transport network. While Bandon is connected to Cork City and surrounding settlements through regional bus services, the extent to which those services provide an effective and equitable alternative to private-car travel varies across the town. Accessibility is therefore considered not only in terms of the physical location of bus stops, but also in relation to the characteristics and needs of the population living within different parts of Bandon.

Research Questions

Is public transport a viable alternative to private-car commuting between Bandon and Cork City?

Which areas are potentially most vulnerable to limited public transport accessibility because of their demographic, health, employment, or household characteristics?

The first question focuses on the spatial performance of the public transport system through walking accessibility and service frequency. The second recognises that accessibility should be evaluated within a broader social context, acknowledging that some population groups are more likely to depend on public transport than others. A bus stop may be geographically accessible, yet still fail to provide meaningful accessibility if services are infrequent or if populations with greater mobility needs remain poorly served.

Executive Summary

The hotspot analysis demonstrates that public transport accessibility in Bandon is spatially uneven and strongly centralised. Areas surrounding the town centre benefit from the highest levels of accessibility due to their proximity to the four principal bus stops and the concentration of scheduled bus services. In contrast, several peripheral residential neighbourhoods experience substantially lower accessibility while simultaneously containing populations with characteristics associated with greater transport vulnerability.

The findings indicate that accessibility and vulnerability do not follow the same spatial pattern. Some neighbourhoods have relatively low accessibility despite comparatively low vulnerability, while others combine limited accessibility with demographic and socio-economic characteristics that increase potential dependence on public transport. These latter locations emerge as Critical and High Priority Accessibility-Vulnerability Hotspots.

The analysis provides a spatial evidence base for identifying areas where improvements to walking infrastructure, bus stop provision, service frequency, or broader transport planning interventions could deliver the greatest social benefit.

Answers to the Research Questions

Is public transport a viable alternative to private-car commuting between Bandon and Cork City?

Answer: Partially yes. Public transport is a viable commuting alternative for residents living within the central accessibility corridor where walking access and service availability are strongest. Viability declines across several peripheral residential neighbourhoods where accessibility is lower and walking access is substantially longer.

Which areas of Bandon are potentially most vulnerable to limited public transport accessibility?

Answer: Critical and High Priority Small Areas are located primarily around the outer residential neighbourhoods, where lower accessibility combines with older age profiles, disability or poor health, retirement or unemployment, and households without private-car access.

1. Understanding Public Transport Accessibility

Public transport accessibility describes the ease with which people can reach and use public transport and, through it, access destinations and opportunities. In this study, accessibility is treated as a spatial and multidimensional concept rather than a single distance measure.

1.1 Public Transport Accessibility in Urban and Regional Planning

Accessibility is concerned with the relationship between people, transport systems and opportunities. A location may be physically close to a bus stop but still have relatively weak effective accessibility if services are infrequent. Conversely, a location with a somewhat longer walk may have stronger effective accessibility where frequent services are available. The Bandon analysis therefore separates walking access from service availability before combining them in the composite analysis.

- **Opportunities:** Accessibility links transport provision to access to employment, education, health-care, services and other opportunities.
- **Mobility vs. Accessibility:** Mobility concerns movement itself, whereas accessibility concerns the opportunities that movement enables.
- **Quality of Life:** Accessibility is relevant to quality of life because transport conditions influence people's ability to participate in social and economic life.
- **Equity:** Transport equity requires consideration of both the distribution of service provision and the distribution of population need.

1.2 Accessibility as a Spatial Concept

The Bandon framework calculates accessibility through several related spatial dimensions: distance, time, opportunity, network connectivity and service availability. Walking distance and network travel time describe the physical effort required to reach a stop, while GTFS-derived service frequency introduces a temporal dimension.

- **Distance:** How far a residential area is from a public transport stop through the pedestrian network.
- **Time:** The estimated walking time required to reach the transport network.
- **Opportunity:** The potential to use public transport once a stop is reached.
- **Network Connectivity:** The pedestrian routes available for reaching stops rather than relying solely on straight-line distance.
- **Service Availability:** The frequency and temporal intensity of public transport services derived from GTFS feeds.

1.3 Accessibility in Small Towns

Smaller commuter towns can display a sharp transition between a relatively well-served urban core and lower-density peripheral residential areas. Bandon provides a useful case study because the mapped results show that accessibility is spatially structured rather than uniform across the study boundary.

1.4 Accessibility and Social Equity

Transport disadvantage becomes particularly important where weaker accessibility overlaps with population characteristics associated with greater potential dependence on public transport. The vulnerability layer is therefore not treated as a measure of transport accessibility itself. It is a separate population-need dimension that is combined with accessibility at the hotspot stage.

1.5 Research Context

The Bandon study forms part of the Spatial Evidence Lab portfolio and applies a reproducible GIS approach to a small-town transport question.

Accessibility → Mobility → Opportunity → Quality of Life

2. Study Area and Analytical Framework

The purpose of the study is to assess how public transport accessibility varies spatially across Bandon and to identify locations where lower accessibility coincides with greater population vulnerability.

2.1 Study Area Overview

The study area is defined by the mapped study boundary around Bandon and its surrounding residential and settlement areas. The context map identifies public transport stops and the local road network. The mapped extent captures the town centre and peripheral areas including Gully, Cloughmacsimon, Kilbrogan, Ballylangley, Currough, and Knocknagrohan.

The study uses Small Areas as the principal reporting geography for population vulnerability and spatial comparison. Public transport stops are represented as point features, while the road and pedestrian network provides the spatial context for network-based access analysis.

2.2 Study Context and Population Vulnerability Assessment

Bandon has a resident population of 8,196 (Census 2022) and functions as a key regional settlement and commuter node. Bandon was selected because its defined urban core and surrounding lower-density residential peripheries make it an ideal small-town case study for examining the transition between high and lower public transport accessibility.

Population Vulnerability as a Complementary Dimension

To evaluate transport equity, the analysis incorporates Census 2022 Small Area data to construct a Population Vulnerability Index. This index does not measure accessibility itself; instead, it identifies characteristics that may increase potential dependence on public transport or indicate greater vulnerability to transport disadvantage.

Census Variable	Description	Relevance to Vulnerability Assessment
T1_1AGE70_74T	Population aged 70-74	Identifies older residents who may have increased dependence on accessible transport.
T1_1AGE75_79T	Population aged 75-79	Represents populations with potentially greater mobility needs.
T1_1AGE80_84T	Population aged 80-84	Captures areas with higher concentrations of elderly residents.
T1_1AGEGE_85T	Population aged 85+	Represents population most likely to require highly accessible services.
T1_1AGETT	Total population	Denominator for population-based vulnerability measures.
T4_5RP / T8_1_RT	Retired persons	Represents populations less likely to rely on work travel but dependent on local transport.
T8_1_LTUT	Long-term unemployed	Represents socio-economic vulnerability and transport disadvantage.
T8_1_UTWSDT	Permanent sickness/disability	Represents populations experiencing greater mobility constraints.
T12_1_T	Persons with a disability	Measures populations requiring accessible transport infrastructure.
T12_3_BT	Bad general health	Indicates communities with health-related transport needs.
T12_3_VBT	Very bad general health	Identifies areas with high health-related transport vulnerability.
T15_1_NC	Households without a car	Households potentially most dependent on public transport.
T11_1_BUT	Travelling by bus/minibus	Provides evidence of existing public transport use.
T11_2_T1-T8	Departure time for work/school	Captures commuting demand during morning departure periods.
T11_2_NS / T11_2_T	Leaving time not stated / Total	Denominator for analysing commuting time distributions.

Dimensions of Population Vulnerability

These Census variables are grouped into four core vulnerability domains:

1. Demographic Vulnerability: Older population (70+ years), retired persons.
2. Health and Mobility Vulnerability: Disability, permanent sickness, bad and very bad general health.
3. Socio-Economic Vulnerability: Long-term unemployment and retirement-related indicators.
4. Transport Dependence: Households without a motor car, bus commuters, and morning departure-time patterns.

Integration Framework

The analysis strictly separates the spatial performance of the transport network from population demand and vulnerability:

Census Variable	Purpose in the Study
Public Transport Supply & Accessibility	Measured via bus stop locations, walking-network catchments, and peak service frequency
Observed Travel Behaviour	Represented by Census commuting patterns and bus-use variable
Population Vulnerability	Represented by age, retirement, unemployment, disability, and health indicators
Private Vehicle Availability	Represented by households without access to a motor car

2.3 Research Questions and Objectives

Primary Research Question: **How can GIS-based accessibility and vulnerability analysis identify areas of transport disadvantage in Bandon?**

Research Objectives:

1. Map the spatial coverage of public transport stops using walking-network analysis.
2. Measure walking-distance accessibility to public transport stops.
3. Assess peak-time service availability using GTFS-derived service frequency.
4. Construct a population vulnerability layer using selected Census 2022 indicators.
5. Combine accessibility and vulnerability to identify spatial hotspots and priority areas.

2.4 Analytical Framework and Map Series

Map	Title	Subtitle
01.00	Base Context & Transit Infrastructure	Bandon Built-Up Areas (BUA) and Network Baseline
01.01	Physical Pedestrian Catchments	Network-Based Isochronal Walkability Analysis
01.02	First/Last-Mile Infrastructure Quality	Footpath Coverage and Pedestrian Crossing Networks
01.03	Headway & Operational Frequency	Peak-Period Transit Service Quality
01.04	Demographic Vulnerability & Demand	Spatial Distribution of Priority Transit-Dependent Populations
01.05	Composite Accessibility Index (CAI)	Integrated Spatial-Temporal Transit Performance
01.06	Targeted Intervention Framework	Spatial Decision-Support Priorities for Capital Investment

3. Dimensions and Methodology

The analytical pipeline processes accessibility through progressive dimensions:

Walking Access → Distance Score → Service Frequency → Vulnerability → Integrated Priority

3.1 Walking Accessibility (Map 01.01)

Evaluates 5, 10, and 15 minute network catchments around stops using a standard 5 km/h walking speed. Network routing provides a true representation of pedestrian access over straight-line buffers.

3.2 Distance-Based Accessibility (Map 01.02)

Classifies continuous distance scores into Low, Moderate, High, and Very High accessibility, reflecting physical infrastructure coverage.

3.3 Service Availability (Map 01.03)

Adds scheduled peak-time service frequency (headways) to differentiate physical proximity from actual service utility (Low, Medium, High).

3.4 Population Vulnerability (Map 01.04)

Mapped into five relative classes (Very Low, Low, Moderate, High, Very High) across Small Areas based on demographic transit dependency.

3.5 Composite Accessibility Index (Map 01.05)

Integrates pedestrian accessibility and transit service frequency into a single Composite Accessibility Index (CAI), rated on a five-tier scale (Very Poor, Poor, Moderate, Good, Excellent).

3.6 Equity-Focused Vulnerability Hotspot Model (Map 01.06)

Overlays CAI and Vulnerability to create the decision-support prioritisation matrix:

Priority Category	Spatial & Social Interpretation
Critical Hotspot	Very low accessibility combined with high population vulnerability
High Priority	Low accessibility with elevated population vulnerability
Moderate	Intermediate accessibility and vulnerability scores
Low Priority	Lower vulnerability despite reduced physical accessibility
Well Served	High transport accessibility relative to population vulnerability

3.7 Data Infrastructure and Sources

The study integrates NTA GTFS schedule feeds, CSO Census 2022 Small Area statistics, OpenStreetMap pedestrian networks, and Tailte Éireann boundary data:

Data Source	Contribution to the Framework
GTFS Stops, Routes & Trips (NTA)	Public transport locations, routing, and peak timetable
OpenStreetMap	Pedestrian network routing and street topology
CSO Census 2022	Socio-demographic, health, age, and car-ownership indicators
Small Areas - Tailte Éireann / CSO	Primary spatial reporting geography

3.8 Methodology and Workflow

1. Network Processing: Build a connected pedestrian routing network in GIS.
2. Walking Catchment Generation: Calculate 5, 10, and 15 minute service areas from GTFS stops at 5 km/h.
3. Temporal Frequency Calculation: Extract peak morning trips from GTFS stop times and calculate service frequency scores.
4. Vulnerability Index: Groups census variable into four derived indicators (Elderly, Health, No Car, and Econ and combines them into a composite value.
5. Composite Indexing: Combine walking access scores and service frequency into Composite Accessibility Index (CAI).
6. Equity-Focused Vulnerability Hotspot Model : Matrix overlay of CAI and Vulnerability Index to generate Map 01.06.

3.9 Walking Accessibility Analysis

Walking accessibility is modelled using the pedestrian network rather than straight-line distance. Network service areas are generated around public transport stops and expressed using 5, 10 and 15 minute walking thresholds (assumed walking speed of 5 km/h). A distance-based score is then derived to provide a more continuous accessibility representation.

3.10 Service Frequency Analysis

GTFS schedule data are used to represent the temporal dimension of public transport accessibility. Trips and stop-time information are processed to estimate peak-time service frequency, which is converted into a frequency score and used in the frequency-adjusted accessibility map.

3.11 Vulnerability Index Construction

A Population Vulnerability Index is constructed from selected Census 2022 indicators. The indicators are standardised and combined to create a spatial measure of relative vulnerability. The resulting geography is classified into five levels from Very Low to Very High.

3.12 Composite Accessibility Index

The Composite Accessibility Index integrates the principal accessibility dimensions represented by walking access and service frequency. The resulting CAI is classified into five levels: Very Poor, Poor, Moderate, Good and Excellent.

3.13 Hotspot Analysis and Equity-Led Prioritisation Framework

The Equity-Focused Vulnerability Hotspot Model (Map 01.06) identifies priority areas by integrating physical transport accessibility deficits with socio-demographic vulnerability. Rather than evaluating physical access in isolation, this approach adopts an equity-led prioritisation framework to ensure that public transport and pedestrian infrastructure investments target communities with the greatest potential transport dependency.

- **Core Model Variables** = 1 - Normalised CAI. Higher values denote poorer transit accessibility and longer network walking distances. VUL_norm represents composite population vulnerability.
- **Mathematical Formulation & Weighting Schema:** Hotspot Score Equity = $0.4 \times \text{Access Deficit} + 0.6 \times \text{VUL_norm}$
- **Policy Rationale and Spatial Interpretation:** By placing greater weight on population vulnerability (60%), the model prevents high-priority ranks from being assigned solely to unpopulated or low-density peripheral areas. Instead, higher vulnerability exerts a stronger pull on the final score.
- **4. Benchmark Comparison:** For sensitivity analysis and policy evaluation, the dataset includes a secondary, balanced benchmark model (50/50 split)

4. Results and Spatial Interpretation

The map series provides an immediate visual reading of the spatial structure of public transport accessibility in Bandon. The most consistent pattern is a stronger accessibility core around the town centre and weaker conditions toward peripheral areas.

4.1 Map 01.00 - Base Context & Transit Infrastructure

SEL - 002 01.00 | BASE CONTEXT & TRANSIT INFRASTRUCTURE

Bandon Built Up Areas (BUA) and Network Baseline



Legend

- Bus stop
- Study area
- Primary roads
- Local streets

Context Map



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The study establishes the spatial context for assessing public transport accessibility in Bandon, showing the study boundary, public transport stops and the local road network that supports movement throughout the town.

DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map establishes the spatial context for the accessibility assessment, showing the Bandon study boundary, public transport stops and the surrounding road network. It identifies the relationship between the built-up settlement, the location of public transport infrastructure and the peripheral residential areas that are subsequently assessed through network-based accessibility analysis.

Key Message

Establishes spatial boundaries, major arterial routes, and the distribution of active public transport stops across the study area.

4.2 Map 01.01 - Physical Pedestrian Catchments

SEL - 002 01.01 | PHYSICAL PEDESTRIAN CATCHMENTS

Network-Based Isochronal Walkability Analysis



Legend

- Bus stop
- Study area
- Pedestrian Catchments**
 - 0 - 5 min
 - 5 - 10 min
 - 10 - 15 min
- Primary roads
- Local streets

Context Map



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

Network-based service areas identify areas reachable from public transport stops within 5, 10 and 15 minutes of walking, using a 5 km/h walking speed. Accessibility decreases with increasing walking time, highlighting areas located farther from public transport stops.

DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map converts network-based walking distance to public transport into a relative accessibility score, classified from Low to Very High. It provides a continuous spatial interpretation of pedestrian access and highlights structural gaps and physical barriers in the footpath network that impede seamless pedestrian access.

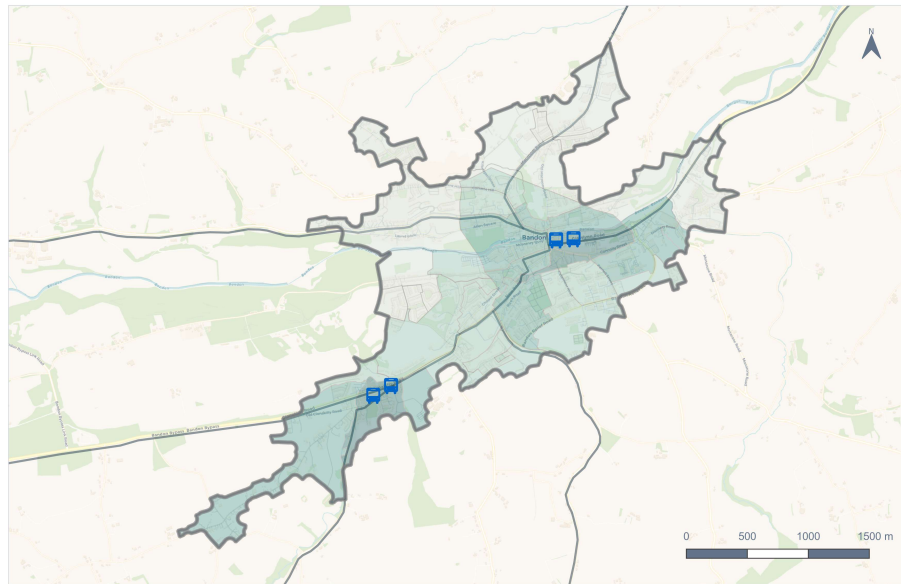
Key Message

Network-based service areas identify areas reachable from public transport stops within 5, 10 and 15 minutes of walking, using a 5 km/h walking speed. Accessibility decreases with increasing walking time, highlighting areas located farther from public transport stops.

4.3 Map 01.02 - First/Last-Mile Infrastructure Quality

SEL - 002 01.02 | FIRST/LAST-MILE INFRASTRUCTURE QUALITY

Footpath Coverage and Pedestrian Crossing Networks



Legend

- Bus stop
- Study area
- Pedestrian Accessibility**
 - Low Accessibility
 - Moderate Accessibility
 - High Accessibility
 - Very High Accessibility
- Primary roads
- Local streets

Context Map



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The study area shows a clear concentration of higher walking accessibility around Bandon's central area and key public transport stops, while accessibility decreases towards the outer parts of the town. This highlights the importance of improving pedestrian connections between residential areas and bus stops to support more equitable access to public transport.

DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map converts network-based walking distance to public transport into a relative accessibility score, classified from Low to Very High. It provides a continuous spatial interpretation of pedestrian access and highlights structural gaps and physical barriers in the footpath network that impede seamless pedestrian access.

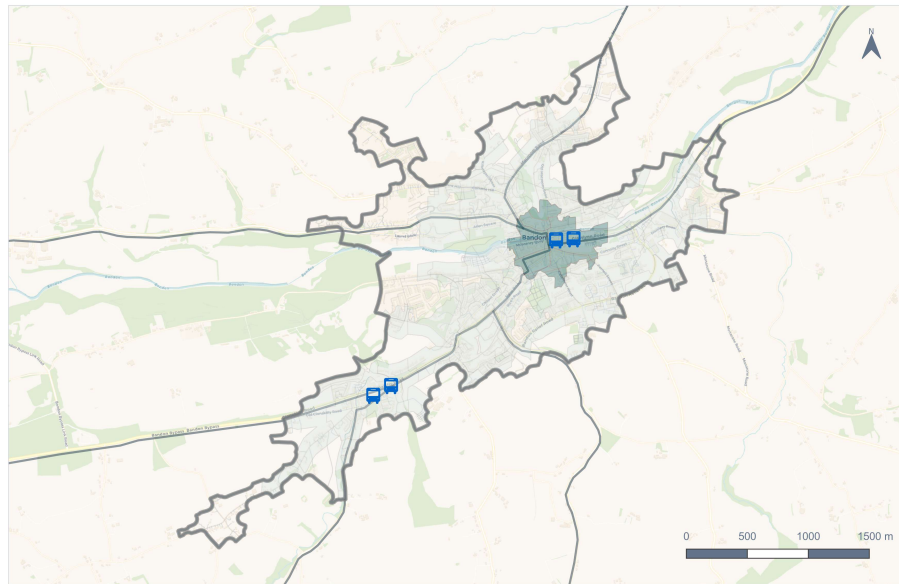
Key Message

Identifies structural gaps and physical barriers in the footpath network that impede seamless pedestrian access to transit stops.

4.4 Map 01.03 - Headway & Operational Frequency

SEL - 002 01.03 | HEADWAY & OPERATIONAL FREQUENCY

Peak-Period Transit Service Quality



Legend

- Bus stop
- Study area
- Peak-Time Service**
 - Low
 - Moderate
 - High
- Primary roads
- Local streets

Context Map



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The map shows that public transport coverage is concentrated around Bandon's central urban area, with limited access in the town's outer areas. This highlights gaps in public transport reach, particularly for communities located away from the main bus routes and stops.

DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map introduces the temporal dimension of accessibility by combining spatial access to public transport with scheduled peak-time service frequency derived from GTFS data. It highlights service quality by mapping peak-hour bus frequencies across existing routes.

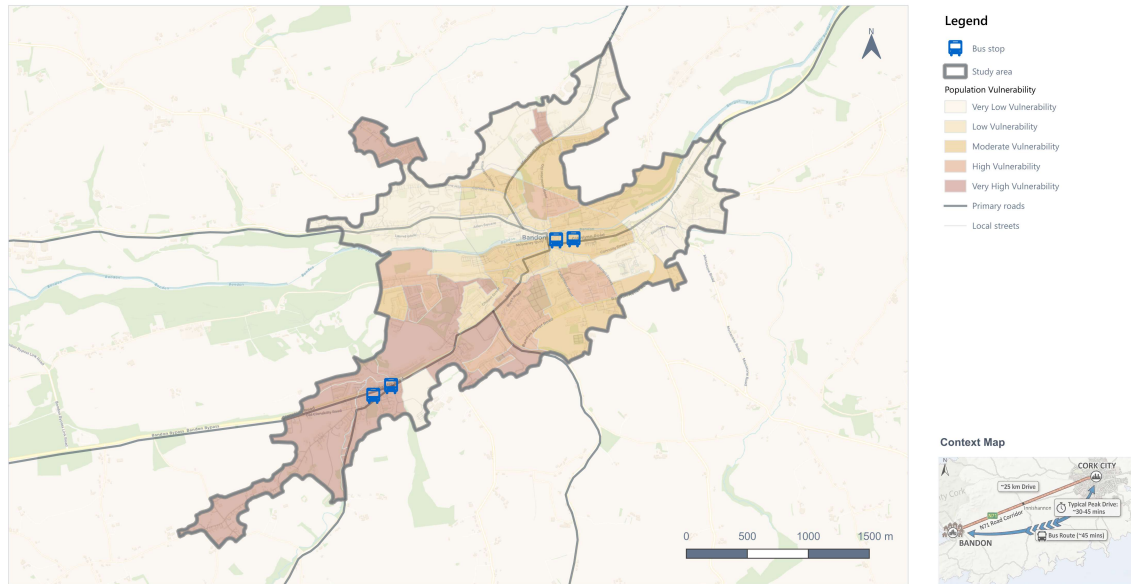
Key Message

Highlights transit service levels by mapping peak-hour bus frequencies and headways across existing routes.

4.5 Map 01.04 - Demographic Vulnerability & Demand

SEL - 002 01.04 | DEMOGRAPHIC VULNERABILITY & DEMAND

Spatial Distribution of Priority Transit-Dependent Populations



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

The map highlights higher population vulnerability across several residential areas, particularly south and southwest of Bandon's centre, where vulnerability overlaps with more limited public transport accessibility. This suggests that older, health- and disability-affected, socio-economically vulnerable and transport-dependent populations may face greater barriers to accessing public transport, reinforcing the need to prioritise improved service coverage and connections in these areas.

DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map represents the population-need dimension of the study using Census 2022 Small Area indicators relating to demographic, health and mobility, socio-economic and transport-dependence characteristics. It pinpoints areas with high concentrations of vulnerable residents requiring reliable transit.

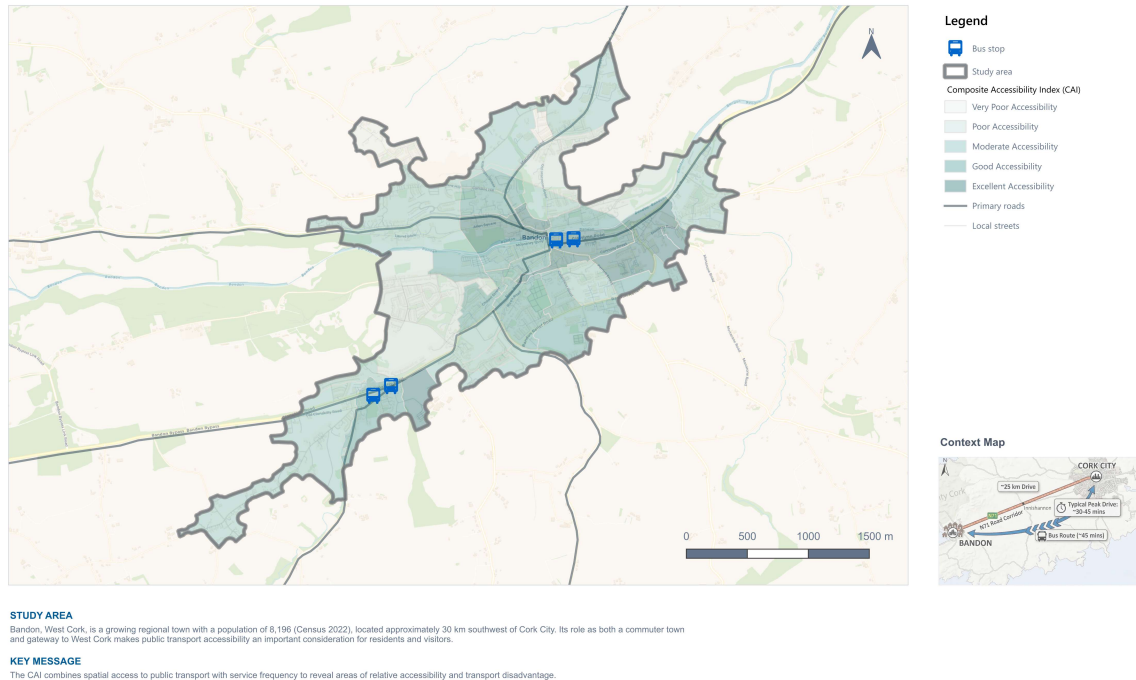
Key Message

Pinpoints areas with high concentrations of vulnerable residents (elderly, low-income, disabled) requiring reliable public transport.

4.6 Map 01.05 - Composite Accessibility Index (CAI)

SEL - 002 01.05 | COMPOSITE ACCESSIBILITY INDEX (CAI)

Integrated Spatial-Temporal Transit Performance



DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

The Composite Accessibility Index (CAI) integrates the two principal transport-access dimensions assessed in the study: network-based walking accessibility and scheduled service frequency. By combining physical access with service availability, the CAI provides an overall spatial hierarchy of public transport accessibility ranging from Very Poor to Excellent.

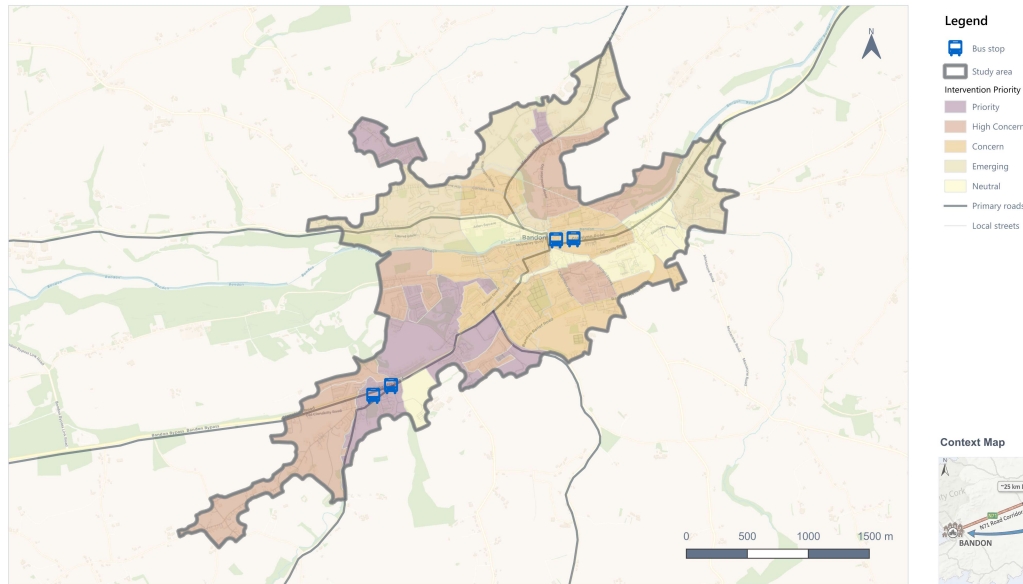
Key Message

Combines physical walkability and service frequency into a single score to highlight spatial disparities in overall public transport provision.

4.7 Map 01.06 - Targeted Intervention Framework

SEL - 002 01.06 | TARGETED INTERVENTION FRAMEWORK

Spatial Decision-Support Priorities for Capital Investment



STUDY AREA

Bandon, West Cork, is a growing regional town with a population of 8,196 (Census 2022), located approximately 30 km southwest of Cork City. Its role as both a commuter town and gateway to West Cork makes public transport accessibility an important consideration for residents and visitors.

KEY MESSAGE

Priority areas emerge where accessibility deficits and population vulnerability overlap, highlighting locations for targeted transport intervention.

DATA SOURCES

CSO - OpenStreetMap - NTA - Tailte Éireann

METHOD & GEOGRAPHY

Accessibility-Vulnerability Spatial Integration Model (AVSIM) - Irish Transverse Mercator (ITM)



Description

This map identifies areas where public transport accessibility conditions intersect with population vulnerability to establish an equity-focused spatial priority framework. Overlays CAI deficits with population vulnerability to delineate high-priority areas for capital investment.

Key Message

Overlays CAI deficits with population vulnerability to delineate high-priority areas for infrastructure upgrades, DRT, or route adjustments.

4.8 Cross-Map Synthesis

Across the seven maps, the evidence develops from network coverage to spatial accessibility, service availability, population vulnerability and finally combined priority. The consistency of the mapped core-periphery pattern strengthens the interpretation that public transport accessibility in Bandon is spatially uneven.

5. Evaluation of Research Questions

The central analytical finding is that public transport accessibility cannot be adequately represented by a single distance measure. The Bandon map series demonstrates the importance of combining physical access, service availability and population characteristics.

- **Accessibility is More Than Distance:** Walking distance is essential, but frequency-adjusted metrics provide a complete representation of system performance.
- **Walking Accessibility and Urban Form:** The urban core is closely connected to stops, while peripheral residential areas fall into weaker accessibility classes.
- **Temporal Accessibility:** Peak-time frequency introduces time into the accessibility framework.
- **Population Need vs. Service Provision:** Comparing vulnerability with CAI distinguishes poorly served but less vulnerable areas from high-need transport cold spots.
- **Composite Accessibility Index:** The CAI serves as the principal integrated accessibility metric in the study.
- **Accessibility-Vulnerability Hotspots:** The hotspot layer moves from measurement to actionable decision-support prioritisation.

5.1 Research Question One - Viability for Cork City Commuters

Executive Finding: Public transport is a viable alternative for residents in the central corridor, but significantly less viable for peripheral neighbourhoods:

Spatial Zone	Viability Assessment	Key Factors
Central Corridor	High	Short walking distance, high peak service frequency, direct regional bus runs to Cork City
Intermediate Areas	Moderate	Reasonable walking access, but declining peak service availability
Peripheral Areas	Limited / Low	Long walking journeys, reduced service coverage, making private cars far more competitive

5.2 Research Question Two - Identifying Vulnerable Areas and Groups

Executive Finding: Several peripheral neighbourhoods combine low public transport accessibility with higher population vulnerability, marking them as key areas of transport disadvantage.

Affected Group	Primary Barrier	Planning Need
Older Adults (70+)	Reduced walking radius to bus stops	Highly accessible, localized stop placement
Disabled / Poor Health	Physical network barriers and topography	High-quality footpaths, crossings, and low-floor buses
Long-Term Unemployed	High transport costs or poor access to job centres	Affordable, frequent peak commuting services
Car-Less Households	Total reliance on active travel and public transit	High-density transit coverage in residential sectors

5.3 Spatial Typology of Bandon

- **System 1 (High Access / Lower Vulnerability):** Town core; well served by existing regional bus lines.

- **System 2 (Moderate Access / Mixed Vulnerability):** Transitional residential zones requiring selective service adjustments.
- **System 3 (Low Access / High Vulnerability):** Critical and High Priority Hotspots in peripheral estates where transport disadvantage is concentrated.

6. Policy and Planning Implications

6.1 Priority Intervention Areas

Critical Hotspot and High Priority areas mapped in 01.06 should be considered candidate locations for immediate capital assessment.

6.2 Pedestrian Infrastructure Improvements

Where poor accessibility is driven primarily by walking access (Map 01.02), potential interventions include improved footpaths, safe pedestrian crossings, and removal of physical barriers along key stop access corridors.

6.3 Service Frequency Optimisation

Where walking access is reasonable but peak-time service is weaker (Map 01.03), service-frequency interventions and headway adjustments offer a better response than civil infrastructure alone.

6.4 Micro-Mobility & Demand-Responsive Transit (DRT)

Peripheral areas outside fixed-route catchments warrant assessment for Demand-Responsive Transit (DRT) or micro-mobility feeder solutions where traditional fixed-route extensions are economically inefficient.

6.5 Accessibility Monitoring

The GIS workflow can be converted into a repeatable monitoring framework, re-running metrics as GTFS schedules, population figures, and pedestrian infrastructure evolve.

6.6 Recommendations for Cork County Council and NTANTA

- Use Map 01.06 as an explicit spatial decision-support framework for regional transport funding allocation.
- Prioritise first/last-mile active travel links connecting peripheral residential clusters in the south and southwest to central corridors.
- Review peak timetable headways at peripheral stops with high population vulnerability.
- Integrate scheduled-service accessibility analysis with wider active-travel and transport-demand evidence.

7. Limitations and Future Research

7.1 Limitations

- **Walking Assumptions:** Network routing assumes a standard 5 km/h walking speed and complete street topology.
- **GTFS Limitations:** GTFS reflects scheduled timetable data rather than real-time vehicle locations or operational delays.
- **Census Data:** Population indicators rely on static Census 2022 Small Area boundaries.
- **Demand Dynamics:** Accessibility measures potential supply and spatial reach rather than observed personal transit trip demand.

7.2 Future Research Priorities

- **GTFS - Real-time Accessibility:** Integrate GTFS-Realtime data to evaluate reliability and delay impacts.
- **Temporal Expansion:** Evaluate off-peak, evening, and weekend accessibility patterns.
- **Accessibility for Schools and Healthcare:** Measure accessibility to specific essential destinations and examine whether vulnerable populations have adequate access to them.
- **Accessibility for Active Travel:** Integrate walking and cycling infrastructure quality, safety and connectivity into the accessibility framework.
- **Integration into Ireland Quality of Life Index:** Develop transport accessibility as a spatial quality-of-life dimension that can be integrated with the wider Ireland Small Area Quality of Life Index framework.
- **Regional Replication:** Replicate workflow across comparable Irish regional towns (e.g. Clonakilty, Skibbereen, Bantry, Midleton, and Mallow).

8. Conclusion

The Bandon Public Transport Accessibility Analysis demonstrates how GIS can integrate transport network accessibility, scheduled service frequency and population vulnerability into a coherent small-area evidence framework.

8.1 Summary of Findings

Public transport accessibility across Bandon is highly centralized. The integration of spatial walkability, temporal service frequency, and demographic vulnerability provides a clear evidence base demonstrating that transport disadvantage is concentrated in southern and southwestern peripheral neighbourhoods.

8.2 Research Contribution

The principal contribution is methodological: the study demonstrates a reproducible spatial workflow for assessing public transport accessibility in a smaller Irish town using openly available spatial data, GTFS schedules and Census 2022 Small Area information. It also provides a framework for linking transport accessibility to wider social-equity and quality-of-life analysis.

8.3 Final Reflection

Ensuring equitable access requires moving beyond simple distance metrics. By placing population need at the centre of transport evaluation, planners can deliver targeted investments that maximize social benefit.

References

- Accessibility Theory: Hansen (1959); Geurs & van Wee (2004)
- Transport Equity: Lucas (2012); Martens (2016)
- GIS Accessibility: ESRI Network Analyst Literature & OpenStreetMap Routing Protocols
- Irish Transport Policy: National Transport Authority (NTA); Department of Transport
- Census & Spatial Analysis: Central Statistics Office (CSO) Census 2022

Appendix A. Data Sources and Reproducibility Register

This appendix records the principal datasets represented in the current study. Exact dataset versions and extraction dates should be added when the production metadata is finalised.

Dataset	Version / Date	Purpose
CSO Census 2022 Small Areas	2022	Population and socio-demographic indicators.
Bus Éireann / NTA GTFS	GTFS feed 2026	Stops, routes, and stop times.
OpenStreetMap	Extract 2026	Walking street network and local spatial context.
Administrative / Small Area Boundaries	Tailte Éireann / CSO	Study and reporting geography.
Spatial Evidence Lab PostGIS Database	Version 1.0	Processed accessibility and vulnerability layers.

Appendix B. Map Production Register

The following register is the authoritative map naming structure for the SEL-002 deliverables.

Map	Title	Subtitle	File Name
01.00	Base Context & Transit Infrastructure	Bandon Built-Up Areas (BUA) and Network Baseline	SEL-002_01.00
01.01	Physical Pedestrian Catchments	Network-Based Isochronal Walkability Analysis	SEL-002_01.01
01.02	First/Last-Mile Infrastructure Quality	Footpath Coverage and Pedestrian Crossing Networks	SEL-002_01.02
01.03	Headway & Operational Frequency	Peak-Period Transit Service Quality	SEL-002_01.03
01.04	Demographic Vulnerability & Demand	Spatial Distribution of Priority Transit-Dependent Populations	SEL-002_01.04
01.05	Composite Accessibility Index (CAI)	Integrated Spatial-Temporal Transit Performance	SEL-002_01.05
01.06	Targeted Intervention Framework	Spatial Decision-Support Priorities for Capital Investment	SEL-002_01.06

Document Control

Project: Project: SEL-002 - Bandon Public Transport Accessibility Analysis

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